

SABIN PYAKUREL

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PROFILE

Full-stack engineer specializing in deploying AI features into production systems using Python and modern JavaScript frameworks. Experienced in building intelligent systems using NLP embeddings, LLM-powered assessment platforms, and workflow automation. Comfortable shipping end-to-end features, working with teams to produce test evaluated systems. Extremely adaptable to new technological changes.

EDUCATION

Bachelor of Computer Engineering
Tribhuvan University

CERTIFICATIONS & BADGES

AWS Knowledge: Serverless | *Amazon Web Services Training Badge*

Learned Lambda and how to implement it in production with timeouts, memory tuning, concurrency limits, reserved concurrency, dead-letter queues, and idempotency for event sources.

PROJECTS

Intelligent ATS & Retrieval-Augmented Classification System | *Django, Python, NLP, Embeddings*

- Built an LLM-powered resume intelligence system using Word2Vec embeddings for semantic search. Across 8 job categories (e.g., backend, ML engineer), achieved 80% classification accuracy using a manually trained Random Forest on cosine similarity features, a 3x improvement over a linear SVM baseline on the same embedding features.
- Implemented GPT-2 perplexity scoring to automatically flag AI-generated and TF-IDF to identify keyword stuffed resumes, reducing manual screening effort by an estimated 60% for borderline submissions over 100 samples.
- Trained Named-Entity Recognition to properly extract skills and present it in user-friendly manner.

Proctored Technical Assessment Platform | *Transformers, Docker, Test-case, Celery, Redis*

- Programmed multi-layered assessment platform (MCQ, coding, subjective, aptitude) with dynamic question delivery, supporting 4 distinct evaluation pipelines within a single unified system.
- Integrated Flan-T5-large LLM for automated reference answer generation using few-shot prompting with constrained output, fully automating and reducing answer evaluation time for recruiters.
- Implemented identity verification using dlib facial recognition (68-point landmarks, ResNet embeddings), maintaining under 5% false-acceptance rate across test sessions.
- Used Docker-based sandboxing to securely execute and evaluate code submissions by creating test-cases in isolated environments, eliminating host-system exposure across all candidate runs.

Course-Grounded AI PDF Study Assistant | *FastAPI, PostgreSQL, ChromaDB, LangChain, RAG*

- Built test safe deployed AI study platform that ingests course PDFs, lets students select exact chapters for RAG chunking, and generates page-level summaries, reducing off-syllabus answers by an estimated 70% across exam-prep sessions.
- Engineered follow up chat sessions about the required chapters for in-depth but course grounded study which helped students prepare wisely than broad.

ADDITIONAL PROJECTS

Unity Game Projects | *Unity, C#*

- Built interactive 2D game systems with UI flow, gameplay loops, and state management.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C#, SQL

Backend / Systems: Django, Django REST Framework, FastAPI, SQLAlchemy, PostgreSQL, Celery, Redis, Docker, REST API design

Frontend / Full-stack: React, HTML, CSS, JavaScript

GenAI / LLM: LangChain, ChromaDB, RAG, Groq, llama-3.1-8b-instant, Word2Vec, GPT-2, Flan-T5, Transformers, Embeddings, Prompt Engineering

ML / NLP: Sentence Transformers, all-MiniLM-L6-v2, TF-IDF, SVM, NLTK, scikit-learn, NumPy, pandas, dlib

Tools & Cloud: Git, GitHub, FFmpeg, PyMuPDF, typesettract OCR, Jinja2, AWS (Serverless), Render, Windows